Michael Bautista

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Embedded Systems Engineer

Motivated Computer Engineer with proficiency in Python, C++, Java, and electronic hardware design. Experienced in embedded systems and seeking opportunities to contribute to PCB design and embedded systems projects.

WORK EXPERIENCE

Electronic Engineer | Wintriss Controls Group | Acton, MA | Contract

- Conducted expert technical support for over 60,000 electronic units, specializing in root cause analysis.
- Ensured optimal functionality through comprehensive testing, calibration, routine maintenance and validation, utilizing oscilloscope, multimeter, and simulation equipment.
- Proficient in interpreting and creating circuit schematics for various electronic systems.
- Applied strong analytical skills to identify and efficiently resolve complex electronic issues. Collaborated with cross-functional teams.

Help Desk Technician | Worcester Technical High School | Worcester, MA

- Maintained desktops, laptops, printers, workstations, website, active directory etc. for over 1000 students, 100 faculty members and 22 different departments
- Assisted with projects such as digital signage & wall mounted smartboard/projector installation
- Responded and resolved tickets in a timely manner
- Assisted with projects such as digital signage & wall mounted smartboard/projector installation

SEP 2015 – MAY 2019	SEP 2019 – FEB 2023	DEC 2023 – JUNE 2024
Worcester Tech	UMass Lowell	Wintriss Controls Group
IT Help desk	Computer Engineering	Electrical Engineer
Vocational experience	Academic experience	Work experience

EDUCATION

Bachelor of Science in Computer Engineering

University Of Massachusetts | Lowell, MA, USA | GPA: 3.319 | Jan 2019 - Feb 2023

- Graduated cum laude
- Dean’s List

PROJECTS

Canal cleaning unmanned surface vehicle

May 2022 – Feb 2023

- Added Unmanned Surface Vehicle hardware and software configuration to motor controller, gyroscope and ultrasonics on a Raspberry pi 4 utilizing ROS
- Created schematic for current state of USV using CAD software for design of electronic hardware
- Left well documented code and handoff document to future team members

Image Enhancement and Filtering (using C)

May 2022 – Feb 2023

- Implemented Gaussian and Butterworth filtering for noise reduction and image quality improvement
- Developed edge detection using Sobel and Canny kernels for feature extraction and object detection
- Optimized filter parameters through testing and validation for enhanced image results.

Voice Activated Fireplace

May 2021 – Dec 2021

- Developed a Microphone Controlled Fireplace circuit using D-type flip-flops, 555 timers, LEDs, etc…
- Enabled sound activation using clapping sounds, demonstrating innovative use of audio input for control systems

SKILLS

- Programming: Python (2 years), C++ (8 years), Java (1.5 years), VHDL (0.5 years), C (8 years)
- Hardware Design: Digital Logic, Circuit Design, Multisim, Quartus, KiCAD
- Software Development: Eclipse, MATLAB, Visual Studio, Linux, Debugging, Digital Image Processing, Embedded Software/Processing, Raspberry Pi, ROS, Socket Programming, I2C, UART, FPGA, PIC32, PIC24, ALTMEGA 2560, STM32
- Electronics: Multimeter, Oscilloscope, Function Generator, SMD soldering, Through-hole soldering, Desoldering